ASA INTERNATIONAL DATAQUEST 2026 · TÜRKİYE

Already Baked In

Splitting every country's population change to 2100 into momentum, mortality, fertility and migration

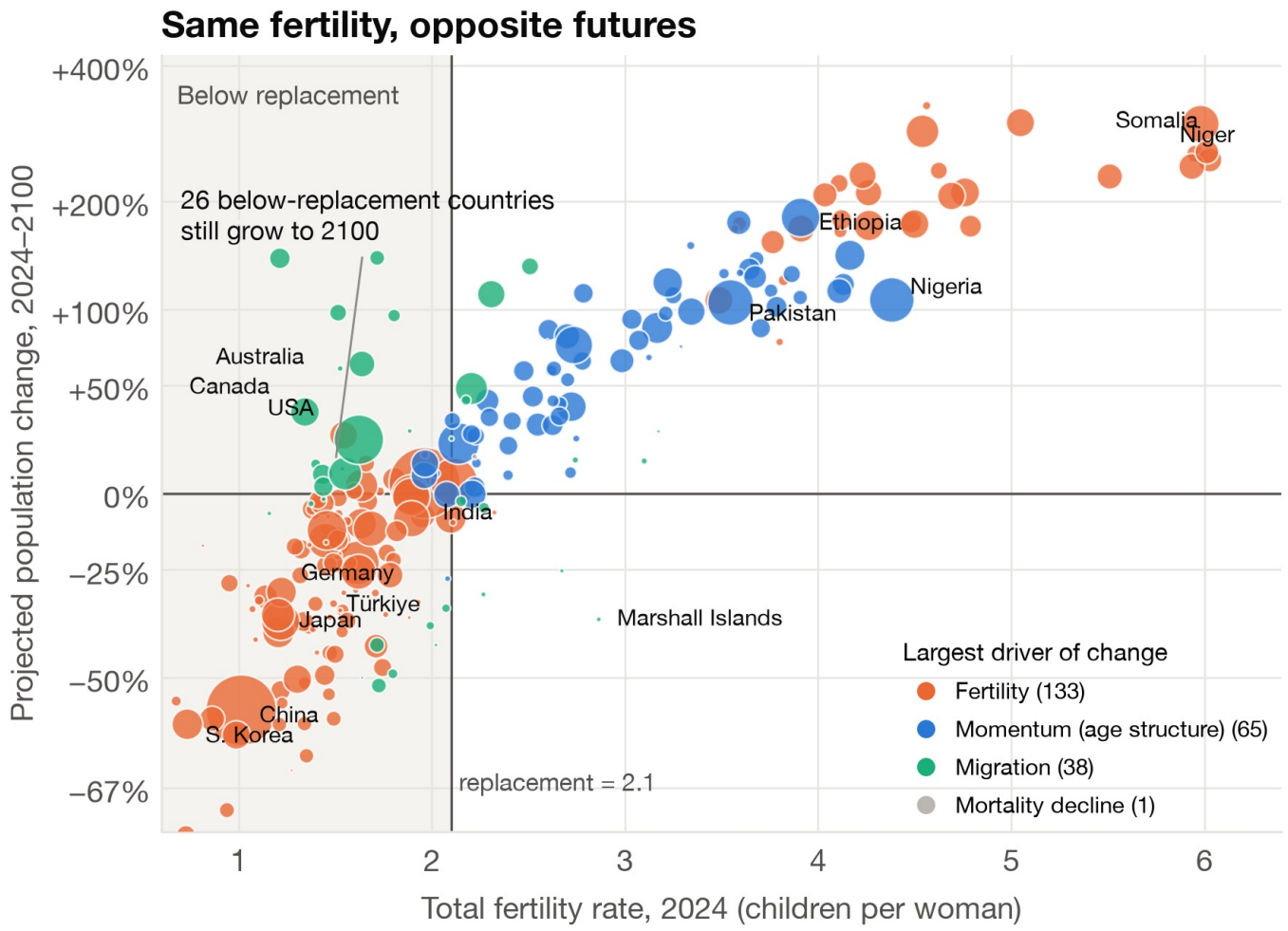


World population, UN medium projection, decomposed with the UN's own counterfactual projection variants

Team: [Team member names] · [University] · [Undergraduate / Master's]

THE PUZZLE

A fertility rate alone does not tell a country's future



Research question

How do momentum, mortality, fertility and migration combine to move each country's population from 2024 to 2100 — and is any single indicator enough to tell the story?

131 countries and areas had below-replacement fertility (TFR < 2.1) in 2024

51 of them — 2.6 bn people — still have age structures that push growth

26 still grow by 2100: India, USA, UK, France, Canada…

Data: 237 countries and areas, 1950–2100, plus 17 alternative UN projection variants. 318 aggregate rows removed (summing every row counts the world 44.8 times); country totals match the UN World row to within 3.6×10^{-7}.

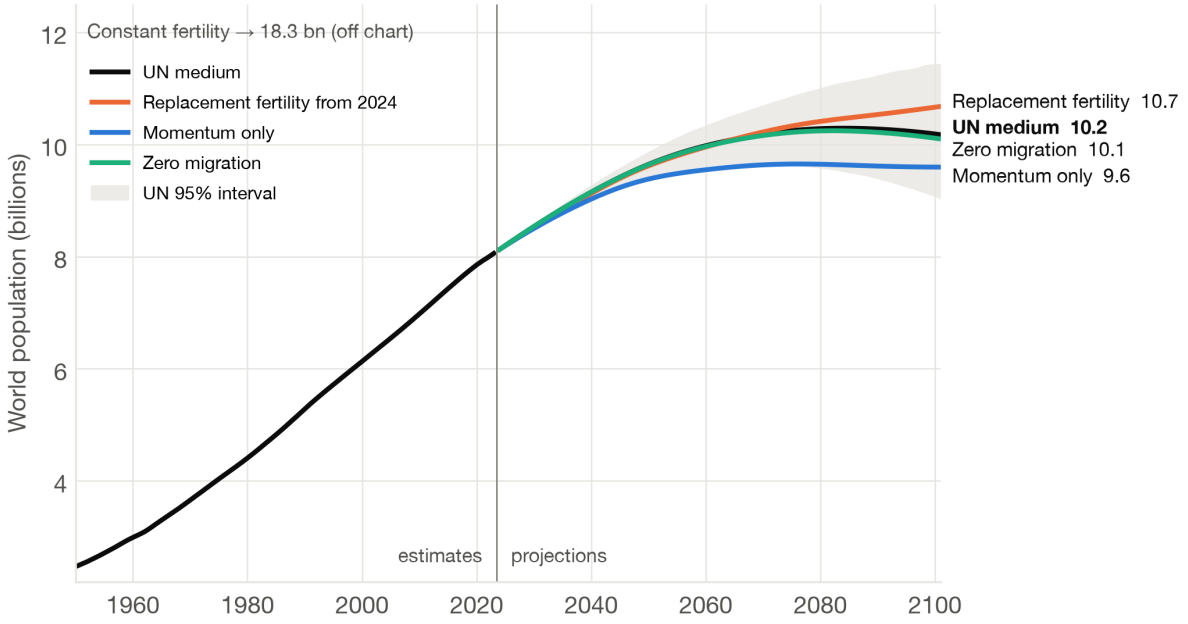
METHOD

Four UN scenario runs split growth into four exact factors

Start: population on 1 Jan 2024	**8.13 bn**
↓ **Momentum ×1.18**	
Momentum run: replacement fertility, constant mortality, no migration	**9.60 bn**
↓ **Mortality ×1.11**	
+ projected mortality decline (instant-replacement, zero-migration run)	**10.68 bn**
↓ **Fertility ×0.95**	
+ projected fertility (zero-migration run)	**10.10 bn**
↓ **Migration ×1.01**	
+ projected migration = UN medium projection	**10.17 bn**

P_{2100} / P_{2024} = momentum × mortality × fertility × migration
Each factor is a ratio of two runs, so the product is exact — no residual. In logs the four effects add up (Andreev, Kantorová & Bongaarts, UN 2013). Same chain for all 237 countries; world = sum of countries.

Even 2.1 children per woman from today ends above the UN path

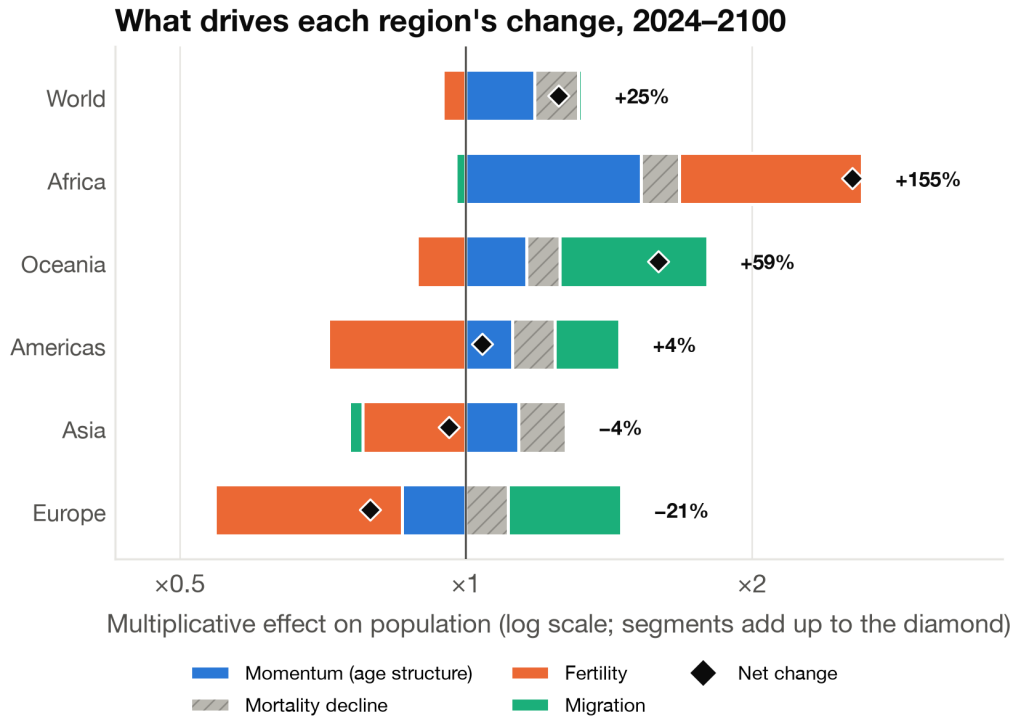


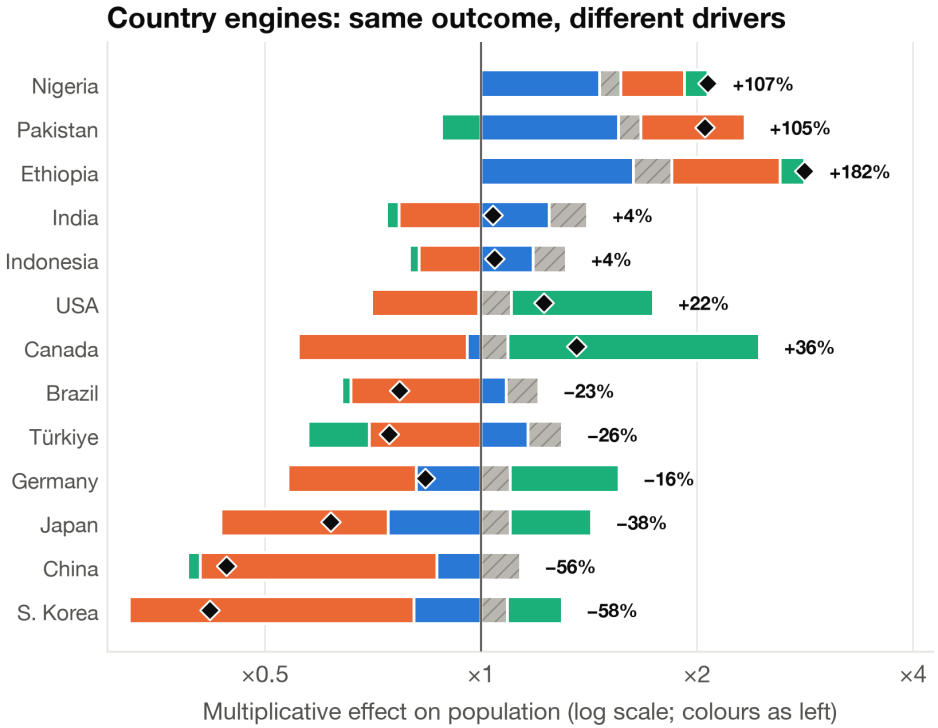
Fertility already pulls the world down

Fertility is a brake on world growth (×0.95). The extra 2.0 bn to 2100 come from today's young age structure (×1.18) and longer lives (×1.11). Even 2.1 children per woman from 2024 would end at 10.7 bn — above the UN path.

FINDINGS

Same outcome, different engines





What separates countries?

Share of the cross-country variance in growth:

fertility 59%

momentum 32%

migration 9%

mortality ≈ 0

Africa ×2.55

A young age structure (×1.53) and above-replacement fertility (×1.56) compound.

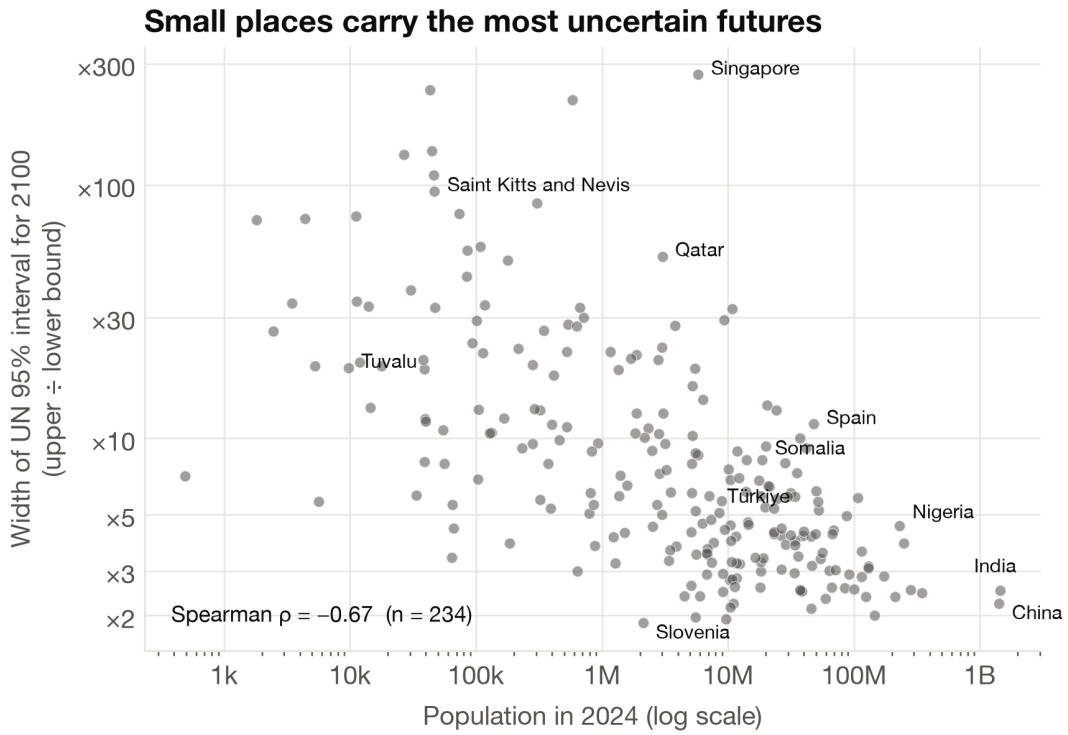
Europe ×0.79

Migration (×1.31) is the main force pushing up; without it Europe would change by −40%.

Robust to the method

Re-estimating each factor along a second chain of UN runs: r = 0.99 (fertility), 0.99 (migration); the largest driver is unchanged for 97% of countries.

Read these numbers as scenarios, not destinies



The UN 95% interval for 2100 spans a factor of ×5.9 for the median country (world: 9.0–11.4 bn). Its width tracks population size (ρ = −0.67), not which engine drives growth (|ρ| ≤ 0.2). For Andorra, Hong Kong and Kuwait the lower bound is exactly 0 — a model artefact we flag rather than hide.

Our ethical choices, following ASA's Ethical Guidelines (2022)

B **Projections are not data**
Every post-2023 value is a model scenario. We show UN intervals, state the chain-order assumption, and test it.

C **Counterfactuals are not targets**
“Zero migration” and “replacement fertility” are accounting devices, not policy goals; childbearing is an individual right.

D **Respect the people counted**
Migration turns decline into growth for 21 countries. We report it descriptively and avoid “replacement” or crisis framing.

B **Weighting is a choice**
Counting countries treats Tuvalu like India. Weighted by population, migration's share of the variance falls from 9% to 1% — so we report both.

A **Open and reproducible**
All code is public; each number is regenerated from the raw UN files with built-in checks.

Letters = ASA principles: A integrity & accountability · B integrity of data & methods · C stakeholders · D those affected. UN location names are used as published and imply no position on legal status.

REFERENCES

Sources and code

1. United Nations, Department of Economic and Social Affairs, Population Division (2024). World Population Prospects 2024, Online Edition. Demographic Indicators (Medium and other variants) and supplement on births by sex and life expectancy at 60. CC BY 3.0 IGO.
2. Andreev, K., Kantorová, V. & Bongaarts, J. (2013). Demographic components of future population growth. UN Population Division, Technical Paper No. 2013/3.
3. Bongaarts, J. (1994). Population policy options in the developing world. Science, 263(5148), 771–776.
4. Keyfitz, N. (1971). On the momentum of population growth. Demography, 8(1), 71–80.
5. Raftery, A. E., Li, N., Ševčíková, H., Gerland, P. & Heilig, G. K. (2012). Bayesian probabilistic population projections for all countries. PNAS, 109(35), 13915–13921.
6. American Statistical Association (2022). Ethical Guidelines for Statistical Practice.
7. Code and data pipeline: github.com/idiltuzkaya/asa_dataquest (cleaning, decomposition, figures; all numbers regenerate from the raw UN files).